

11 Ans: B

$$r_{\text{new}} = 300 \text{ mm}$$

$$r_{\text{worn}} = 294 \text{ mm}$$

Let's say that $T = 2\text{s}$, then $\omega = 2\pi/2 = \pi \text{ rad s}^{-1}$

If the angular velocity is the same as the new (larger) tyre and the worn (smaller) tyre, the speedometer will show the speed based on the calibration of the new (larger) tyre.

$$v_{\text{shown}} = r_{\text{new}}\omega = (300\text{mm})(\pi)$$

$$v_{\text{worn}} = r_{\text{worn}}\omega = (294\text{mm})(\pi)$$

$$\text{percentage difference} = \frac{|(300\text{mm})(\pi) - (294\text{mm})(\pi)|}{(294\text{mm})(\pi)} \times 100\% = 2.04\%$$